Moreau Envelope

$$f_\mu(x) = \min \frac{1}{2\mu} \| x - z \|^2 + f(z)$$

Proximal Operator

$$\mathrm{Prox}_f^\mu(x) = \arg\min \frac{1}{2\mu} \| x - z \|^2 + f(z)$$

若 f 为 L_0-cnt，则 f_μ 也 L_0-cnt（和 μ 无关）

$$| f_\mu(z_1) - f_\mu(z_2) | \leq L_0 \| z_1 - z_2 \|$$

$$\| \nabla f_\mu \| \leq L_0$$ Pf：

若 f 为 convex，则 ∇f_μ 为 $\frac{1}{\mu}$-smooth

$$\| \nabla f_\mu(z_1) - \nabla f_\mu(z_2) \| \leq L_{f_\mu} \| z_1 - z_2 \|$$

Pf：Lemma 3.1 in "Variable Smoothing ..."

$$\nabla H_\mu(w) = \frac{1}{\mu}(w - \mathrm{prox}_g^\mu(w))$$

$$\| \nabla H_\mu(w) - \nabla H_\mu(v) \| = \frac{1}{\mu} \| w - v - (\mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v)) \|$$

由于 g 凸，$\langle w - v, \mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v) \rangle$

$$\geq \| \mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v) \|^2$$

故 $\| w - v - (\mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v)) \|^2$

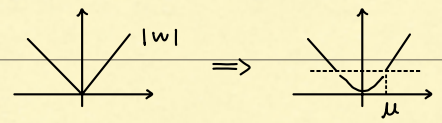
$$\leq \| w - v \|^2 - \| \mathrm{prox}_g^\mu(w) - \mathrm{prox}_g^\mu(v) \|^2 \leq \| w - v \|^2$$

即 $\| \nabla H_\mu(w) - \nabla H_\mu(v) \| \leq \frac{1}{\mu} \| w - v \|$

$$\nabla f_\mu(z) = \frac{1}{\mu} \left[z - \mathrm{Prox}_f^\mu(z) \right]$$ Pf：

[例] $h(w) = |w|$

$$h_\mu(w) = \inf_v \frac{1}{2\mu}|w-v|^2 + h(w) = \begin{cases} |w| - \frac{\mu}{2}, & |w| \geq \mu \\ \frac{|w|^2}{2\mu}, & |w| < \mu \end{cases}$$



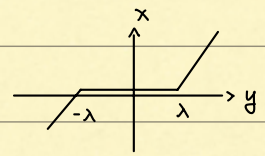
Huber function

$$\mathrm{Prox}_h^\mu(w) = \arg\min_v \frac{1}{2\mu}|w-v|^2 + h(w)$$

$$= S_\mu(w)$$

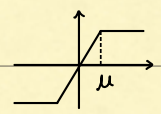
$$= \max\{0,\ x - \mu\,\mathrm{sgn}(x)\} = \max\{0,\ 1 - \frac{\mu}{|x|}\}\cdot x$$

$$= \begin{cases} x-\mu, & x > \mu \\ 0, & -\mu < x \leq \mu \\ -x+\mu, & x \leq -\mu \end{cases}$$



$$\nabla h_\mu(w) = \frac{\partial}{\partial \bar{w}} h_\mu(w) = \begin{cases} \mathrm{sgn}(w), & |w| \geq \mu \\ \frac{w}{\mu}, & |w| < \mu \end{cases}$$

$$L_{\nabla h_\mu} = \frac{1}{\mu}$$



[例] $H(w) = \|w\|_1 = \sum_{m=1}^{M} |w_m|$ Separable

$$H_\mu(w) = \inf_v \frac{1}{2\mu}\|w-v\|^2 + \|w\|_1 = \sum_{m=1}^{M} h_\mu(w_m)$$

可每个分量分别求

$$\mathrm{Prox}_H^\mu(w) = [\,S_\mu(w_m)\,]_{M\times 1}$$

$$\nabla H_\mu(w) = [\,\nabla h_\mu(w_m)\,]_{M\times 1}$$